

EXP : 10

```
import numpy as np
from sklearn.mixture import GaussianMixture

X = np.array([
    [1.0, 2.0],
    [1.5, 1.8],
    [2.0, 2.5],
    [8.0, 8.0],
    [8.5, 9.0],
    [9.0, 8.5]
])

# Create Gaussian Mixture Model
# n_components = number of clusters
model = GaussianMixture(n_components=2, random_state=42)

# Train the model using EM algorithm
model.fit(X)

# Predict cluster for each data point
labels = model.predict(X)

# Display results
print("Data Points and Cluster Labels:")
for i in range(len(X)):
    print(X[i], "-> Cluster", labels[i])

# Display means of the clusters
print("\nCluster Means:")
print(model.means_)
```

```
# Display covariance matrices
print("\nCovariance Matrices:")
print(model.covariances_)
```

```
# Display mixing probabilities
print("\nMixing Probabilities:")
print(model.weights_)
```

```
... Data Points and Cluster Labels:
[1. 2.] -> Cluster 0
[1.5 1.8] -> Cluster 0
[2. 2.5] -> Cluster 0
[8. 8.] -> Cluster 1
[8.5 9. ] -> Cluster 1
[9. 8.5] -> Cluster 1

Cluster Means:
[[1.5 2.1]
 [8.5 8.5]]

Covariance Matrices:
[[[0.16666767 0.08333333]
  [0.08333333 0.08666767]]

  [[0.16666767 0.08333333]
  [0.08333333 0.16666767]]]

Mixing Probabilities:
[0.5 0.5]
```